

Quadcopter UAV Control based on Input-Output Linearization and PID

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Abstract—A dynamical model of the quadcopter UAV is obtained considering the frictional effects in the positional and rotational coordinates. Then, a two-level control strategy is designed using an input-output Linearization in combination with PID controllers. The inner loop stabilizes the pitch, roll and yaw angles, whilst the outer-loop control is related to the position control, avoiding local Linearization. A PID control gain selection is analyzed to harmonize both control loops. The performance of the control approach is shown by numerical simulations.

Nowadays, the Unmanned Aerial Vehicles (UAV's) becomes in disruptive technologies of mobile robotics solutions for surveillance, object transporting, logistics, photography and filming, entertainment, medical aid, among others [1]. The multi-rotor UAV's generate their lift through the forces generated by the propellers of their rotors [2]. A kind of multi-rotor UAV is the named quadcopter UAV, actuated by four out-running motors with propellers mounted at the ends of a cross-shape or plus-shape structure. It becomes popular for their ability to hover over targets, aggressive maneuvers like flip behaviors and agile flight in outdoors and indoors environments [3].

The quadcopter UAV constitutes a nonlinear and underactuated system with omnidirectional motion in the space [4]. The model can be obtained through the newton-Euler, Euler-Lagrange or quaternions methods [5]. A more realistic dynamical model could be complex to the appearance of deflections of propellers, air friction and turbulence, non-symmetric inertia, instant floor-effect in take off, sensor noise, etc [6]. The fundamental control problems include hovering [7], take-off, landing, trajectory tracking, collision avoidance, and recently for logistics [1], swarm with UAV's or ground vehicles, etc.

The rotational dynamics is frequently assumed as pure and decoupled double integrator or including the Coriolis and gravitational matrices [8]. The rotational control is conceived as an inner-loop for the fast convergence of the pitch, roll and yaw [9]. The references signals are provided by a manual control or directly for a positional outer-loop, controlling the (x, y, z) coordinates, for a fully autonomous

flight. This two-level control architecture is inspired by the actual electronics platforms equipped with Inertial Measurement Unit (IMU), vision-based positioning, GPS and flight controllers linked to a radio control for manual control¹.

On the other hand, the positional control is commonly solved by classical controllers applied to linear and reduced dynamical models using Taylor Series [10]. These approaches become simple but limit the control to a neighbor around the equilibria. To avoid this, input-output linearization is designed which produces different control laws according to the order of multiplication of the three rotational matrices. The yaw angle is not controlled as in [11], the pitch and roll are reduced to a double integrator as in [3], the drone is limited to move in a plane only or reducing its omnidirectional abilities to a unicycle-type or differential configuration. The majority of works do not incorporate friction terms in [12]. The positional control requires the localization of the UAV in the space using GPS for outdoors or motion capture systems in indoors.

Inspired in our previous work in [13], the main contributions of this paper are summarized in the next points:

- Obtain a dynamical model with simple friction terms, as the main environmental perturbations. Also, consider the cross-shape configuration instead the plus-shape, preferred by human pilot users.
- Select an appropriated multiplication of the rotational matrices different from [13], and obtain models for the inner and outer loops to avoid local linearization or truncated models.
- Apply PID controllers in both control loops, instead of the PD setup, as in most works as [11].
- Simulate the behavior of the two control-loops harmonizing the settling times. It becomes a practical guide for the control implementation.

I. DYNAMICAL MODELING OF THE QUADCOPTER UAV

Consider the quadcopter UAV given in the Figure 1. Let $\xi = [x, y, z]^T$ and $\eta = [\phi, \theta, \psi]^T$ be the positional and

¹<http://www.proficnc.com/content/13-pixhawk2>

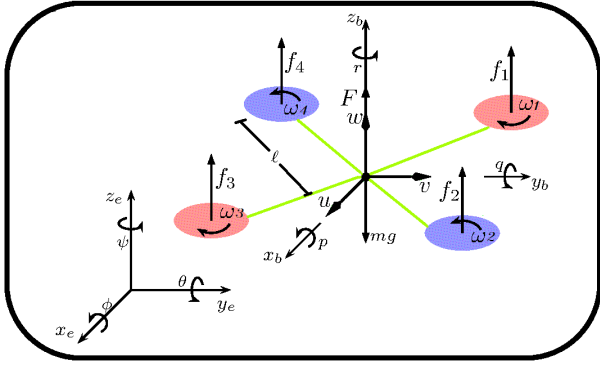


Fig. 1. Quadcopter UAV

rotational coordinates (pitch, roll and yaw), respectively, of the body frame Γ_b respect to fixed frame Γ_e . The linear and angular velocities in Γ_b are $V = [u, v, w]^T$ and $\Omega = [p, q, r]^T$, respectively. The motion in 3D implies three standard rotational matrices given by

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\phi & -s\phi \\ 0 & s\phi & c\phi \end{bmatrix}, R_y = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

$$R_z = \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

where $ca = \cos a$ and $sa = \sin a$, with $a = \phi, \theta, \psi$. The matrices R_x , R_y and R_z do not commute. Therefore, their multiplication given by R_b^e that localise Γ_b respect to Γ_e becomes in different results. In our previous work [13], was chosen $R_b^e = R_x R_y R_z$ similar to [11], which eliminates the ψ angle. A best choice [3] is $R_b^e = R_z R_y R_x$ giving

$$R_b^e = \begin{bmatrix} c\theta c\psi & s\phi s\theta c\psi - c\phi s\psi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\phi s\theta s\psi + c\phi c\psi & c\phi s\theta s\psi + s\phi c\psi \\ -s\theta & s\phi c\theta & c\phi c\theta \end{bmatrix} \quad (1)$$

which includes all the angles. It has demonstrated to be more stable in practical implementations due to the availability of all angles as feedback. The transformation between angular velocities Ω and $\dot{\eta}$ is given by

$$\Omega = W_n \dot{\eta}, \quad \dot{\eta} = W_n^{-1} \Omega, \quad W_n = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & c\theta s\phi \\ 0 & -s\phi & c\theta c\phi \end{bmatrix} \quad (2)$$

A quadcopter vehicle can be conceived as a rigid body with mass and inertia subjected to a gravitational force moving in a 3D space. The total thrust force F in Γ_b generates a vertical force in the axis z_e all time in Γ_e , given by $F_e = R_b^e [0, 0, F]^T$. The dynamical model of the quadcopter UAV studies the behavior of the UAV due to the forces and torques seen from Γ_e . Using the focus of Euler-Lagrange and based in [9], the generalized vector of motion variables is given by $q = [\xi^T, \eta^T]^T$. The Lagrangian $\mathcal{L}(q, \dot{q}) =$

$E_k - E_p$, with E_k and E_p as the kinetic and potential energies, respectively, is given as

$$\mathcal{L}(q, \dot{q}) = \underbrace{\left(\frac{m}{2} \dot{\xi}^T \dot{\xi} + T_{rot} \right)}_{E_k} - \underbrace{[0, 0, mg]^T \xi}_{E_p} \quad (3)$$

where m is the mass in Kg, $g = 9.81 \frac{m}{s^2}$ and $T_{rot} = \frac{1}{2} V^T I V$ in Γ_b , with $I = \text{diag}\{I_{xx}, I_{yy}, I_{zz}\}$ the symmetric matrix of inertia. Substituting (2), $T_{rot} = \frac{1}{2} \dot{\eta}^T J \dot{\eta}$ in Γ_e , with $J(\eta) = W_n^T I W_n$ a symmetric matrix. Thus, the dynamical model can be obtained as

$$\frac{d}{dt} \left[\frac{\partial \mathcal{L}(q, \dot{q})}{\partial \dot{\xi}} \right] - \frac{\partial \mathcal{L}(q, \dot{q})}{\partial \xi} = F_e - A \dot{\xi} \quad (4)$$

$$\frac{d}{dt} \left[\frac{\partial \mathcal{L}(q, \dot{q})}{\partial \dot{\eta}} \right] - \frac{\partial \mathcal{L}(q, \dot{q})}{\partial \eta} = \tau - B \dot{\eta} \quad (5)$$

where $\tau = [\tau_\phi, \tau_\theta, \tau_\psi]^T$ are the torque inputs. The matrices $A = \text{diag}\{A_x, A_y, A_z\}$ and $B = \text{diag}\{b_x, b_y, b_z\}$ are the damping coefficients. $A \dot{\xi}$ represents in a simple manner the air drag or friction for ξ , whilst $B \dot{\eta}$ acts as damping for the rotational motion. Calculating $\frac{d}{dt} \left[\frac{\partial \mathcal{L}(q, \dot{q})}{\partial \dot{\xi}} \right] = m \ddot{\xi}$ and $\frac{\partial \mathcal{L}(q, \dot{q})}{\partial \xi} = -[0, 0, mg]^T$. Also, $\frac{d}{dt} \left[\frac{\partial \mathcal{L}(q, \dot{q})}{\partial \dot{\eta}} \right] = J \ddot{\eta} + \dot{J} \dot{\eta}$. Then, the equations (4)-(5) can be expressed by

$$m \ddot{\xi} + [0, 0, mg]^T = R_b^e [0, 0, 1]^T - A \dot{\xi} \quad (6)$$

$$J \ddot{\eta} + C(\eta, \dot{\eta}) \dot{\eta} = \tau - B \dot{\eta} \quad (7)$$

where the Coriolis matrix is $C(\eta, \dot{\eta}) \dot{\eta} = \dot{J} \dot{\eta} - \frac{1}{2} \frac{\partial}{\partial \eta} (\dot{\eta}^T J \dot{\eta})$. The final model can be written by

$$m \ddot{x} = F (s\phi s\psi + c\phi c\psi s\theta) - A_x \dot{x} \quad (8)$$

$$m \ddot{y} = -F (s\phi c\psi - c\phi s\psi s\theta) - A_y \dot{y} \quad (9)$$

$$m \ddot{z} = F c\phi c\theta - mg - A_z \dot{z} \quad (10)$$

$$J \ddot{\eta} = \tau - C(\eta, \dot{\eta}) \dot{\eta} - B \dot{\eta} \quad (11)$$

Note from Figure 1, that $F = \sum_{i=1}^4 f_i$, where f_i is the individual thrust produced by the i -th motor-propeller array. By experimental experience $f_i = k\omega_i^2 + b_i\omega_i$, with k is the thrust constant, ω_i is the angular velocity of the motor i and the damping $b \approx 0$. It is assumed that all motors and propellers are equals, with no deflecting propellers. For a cross-shape configuration, The desired f_i values can be obtained from F and τ by

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ \delta & \delta & -\delta & -\delta \\ \delta & -\delta & -\delta & \delta \\ \frac{b}{k} & -\frac{b}{k} & \frac{b}{k} & -\frac{b}{k} \end{bmatrix}^{-1} \begin{bmatrix} F \\ \tau \end{bmatrix} \quad (12)$$

with $\delta = \frac{\sqrt{2}}{2} \ell$, and b is the so-called drag constant of the rigid body [8]. the desired angular velocities is obtained from $\omega_i^2 = \sqrt{f_i k^{-1}}$, $i = 1, \dots, n$ and sent to the Electronic Speed Control (ESC) to move the motors. Note in (12), that all motor contributes in the ϕ and θ rotations. This is more stable than the case of plus-shape, where only a pair of motors acts for these angles. The generation of the

resulting torque in (12), is given by the difference of the motors turning clockwise (1 and 3) vs the motors turning counter-clockwise (2 and 4).

II. CONTROL STRATEGY

A. Rotation control

The inner-loop for rotation angles, an input-output Linearization is defined as

$$\tau = J\hat{\tau} + C(\eta, \dot{\eta})\dot{\eta} + B\dot{\eta} \quad (13)$$

which linearize and decouple the equation (11) to $\ddot{\eta} = \hat{\tau}$, with $\hat{\tau}$ as an auxiliary control input. Then, a PID controller for $\hat{\tau}$ is defined as

$$\hat{\tau} = \ddot{\eta}^* - K_d^\eta \dot{e}_\eta - K_p^\eta e_\eta - K_i^\eta \int e_\eta dt \quad (14)$$

where $e_\eta = \eta - \eta^*$ with η^* as the reference values or setpoints for η , $K_d^\eta = \text{diag}\{k_{d\phi}, k_{d\theta}, k_{d\psi}\}$, $K_p^\eta = \text{diag}\{k_{p\phi}, k_{p\theta}, k_{p\psi}\}$ and $K_i^\eta = \text{diag}\{k_{i\phi}, k_{i\theta}, k_{i\psi}\}$, with $k_{d\phi}, k_{d\theta}, k_{d\psi}, k_{p\phi}, k_{p\theta}, k_{p\psi}, k_{i\phi}, k_{i\theta}, k_{i\psi} > 0$. Using (14) it is possible to construct the linear error equation $\ddot{e}_\eta + K_d^\eta \dot{e}_\eta + K_p^\eta e_\eta + K_i^\eta \int e_\eta dt$. Its time derivative is $\ddot{e}_\eta + K_d^\eta \dot{e}_\eta + K_p^\eta e_\eta + K_i^\eta \int e_\eta dt$ and its Laplace transformation is obtained easily as

$$\Phi_\eta(s)E_\eta(s) = 0, \quad \Phi_\eta(s) = s^3 I_3 + s^2 K_d^\eta + s K_p^\eta + K_i^\eta \quad (15)$$

where I_3 is the 3×3 identity matrix. It is clear that the characteristic polynomial $\Phi_\eta(s)$ can be forced to equal to a desired $\Phi_\eta^*(s) = \text{diag}\{\Phi_\phi^*(s), \Phi_\theta^*(s), \Phi_\psi^*(s)\}$, where $\Phi_m^*(s) = (s^2 + 2\zeta^m \omega_n^m s + (\omega_n^m)^2)(s + p^m)$, $m = \phi, \theta, \psi$, where ζ^m is damping coefficient, ω_n^m the natural frequency and p^m is additional non-dominant pole placed in $p_m \geq 10\zeta^m \omega_n^m$. The equality $\Phi_\eta(s) = \Phi_\eta^*(s)$ is achieved if $k_{dm} = 2\zeta^m \omega_n^m + p^m$, $k_{pm} = \omega_n^m + 2\zeta^m \omega_n^m p^m$, $k_{im} = \omega_n^m p^m$. It is useful to adjust ζ^m and ω_n^m according to desired settling time t_{ss}^m by the relation $\zeta^m \omega_n^m = 5(t_{ss}^m)^{-1}$.

B. Position control

To avoid a linear reduced model by Taylor series, we define the total thrust as

$$F = \frac{m(r_z + g) + A_z \dot{z}}{\cos \theta \cos \phi}. \quad (16)$$

where r_z is a new auxiliary control for the axis Z . Note the singularities when $\phi, \theta = \frac{\pi}{2}$. Therefore, it is important to use (16) for $-\frac{\pi}{2} < \phi < \frac{\pi}{2}$ and $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Substituting (16) in (10), it simplifies to $\ddot{z} = r_z$. Also, equations (8) and (9) are modified in matricial form to

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} = \left(r_z + g + \frac{A_z}{m} \right) R(\psi) \begin{bmatrix} \frac{\tan \phi}{\tan \theta} \\ \tan \theta \end{bmatrix} - \begin{bmatrix} \frac{A_x}{m} \dot{x} \\ \frac{A_y}{m} \dot{y} \end{bmatrix} \quad (17)$$

where $R(\psi) = \begin{bmatrix} s\psi & c\psi \\ -c\psi & s\psi \end{bmatrix}$. Since $\det R(\psi) = 1$, it is possible to obtain

$$\begin{bmatrix} \frac{\tan \phi}{\tan \theta} \\ \tan \theta \end{bmatrix} = \frac{1}{r_z + g + \frac{A_z}{m}} R^{-1}(\psi) \begin{bmatrix} r_x + \frac{A_x}{m} \dot{x} \\ r_y + \frac{A_y}{m} \dot{y} \end{bmatrix} \quad (18)$$

where r_x and r_y are the new auxiliary control inputs for the coordinates x and y , respectively. Finally, solving for θ and ϕ , from (18) it gives

$$\theta = \arctan \left[\frac{c\psi \left(r_x + \frac{A_x}{m} \dot{x} \right) + s\psi \left(r_y + \frac{A_y}{m} \dot{y} \right)}{r_z + g + \frac{A_z}{m}} \right] \quad (19)$$

$$\psi = \arctan \left[\frac{c\theta \left[s\psi \left(r_x + \frac{A_x}{m} \dot{x} \right) - c\psi \left(r_y + \frac{A_y}{m} \dot{y} \right) \right]}{r_z + g + \frac{A_z}{m}} \right] \quad (20)$$

The values of the previous equation are the two first references ϕ^* and θ^* of the desired vector η^* of (14) for the rotation control. Similar to (14), auxiliary control inputs can be defined in a matrix manner by

$$r = [r_x, r_y, r_z]^T = \ddot{\xi}^* - K_d^\xi \dot{e}_\xi - K_p^\xi e_\xi - K_i^\xi \int e_\xi dt \quad (21)$$

where $e_\xi = \xi - \xi^*$ with ξ^* as the reference signals for ξ provided by some trajectory generator. The matrices $K_d^\xi = \text{diag}\{k_{dx}, k_{dy}, k_{dz}\}$, $K_p^\xi = \text{diag}\{k_{px}, k_{py}, k_{pz}\}$ and $K_i^\xi = \text{diag}\{k_{ix}, k_{iy}, k_{iz}\}$, with $k_{dx}, k_{dy}, k_{dz}, k_{px}, k_{py}, k_{pz}, k_{ix}, k_{iy}, k_{iz} > 0$. Similar to equation (15), the Laplace transformation is now

$$\Phi_\xi(s)E_\xi(s) = 0, \quad \Phi_\xi(s) = s^3 I_3 + s^2 K_d^\xi + s K_p^\xi + K_i^\xi \quad (22)$$

Consider a $\Phi_\xi^*(s) = \text{diag}\{\Phi_x^*(s), \Phi_y^*(s), \Phi_z^*(s)\}$, with $\Phi_m^*(s) = (s^2 + 2\zeta^m \omega_n^m s + (\omega_n^m)^2)(s + p^m)$, $m = x, y, z$, and $p_m \geq 10\zeta^m \omega_n^m$, then the gain parameters of (14) are adjusted to satisfy $\Phi_\xi(s) = \Phi_\xi^*(s)$ through $k_{dm} = 2\zeta^m \omega_n^m + p^m$, $k_{pm} = \omega_n^m + 2\zeta^m \omega_n^m p^m$, $k_{im} = \omega_n^m p^m$, with ζ^m and ω_n^m obeying a settling time, at least 10 times slower, than the settling time of the inner loop.

Finally, note that the desired reference for the angle ψ is free, but can be obtained selected to follow the velocity vector of x^* and y^* through $\psi^* = \arctan \left(\frac{\dot{y}^*}{\dot{x}^*} \right)$, a constant value or another smooth trajectory.

The final control scheme is presented in the Figure 2, where each box indicates the number of equation obtained for the two control loops. Note the presence of angle limiters in order to avoid the singularities in the control law.

III. NUMERICAL SIMULATION

Figures 3-5 show numerical simulations of the control approach using Matlab-Simulink®. The desired trajectory is a circular path for x and y and a smooth polynomial for z . Thus, $\xi^* = [-3 + 3 \sin(\frac{2\pi}{30}t), 3 \cos(\frac{2\pi}{30}t), \frac{3}{30}t^2 - \frac{2}{30^3}t^3]^T$. The initial values are $\xi(0) = [0, 0, 0]^T$ and $\eta(0) = [0, 0, 0]$. The model parameters are $m = 1$, $A_x = A_y = A_z = 0.01$, $B_\phi = B_\theta = B_\psi = 0.001$, $l = 2$, $k = 0.1$.

For the inner-loop, we set $t_{ss}^\eta = 1$ and $\zeta^\eta = 0.99$ with $p^\phi = p^\theta = p^\psi = 50$, generating the control gains for the PID $k_{p\phi} = k_{p\theta} = k_{p\psi} = 525.5$, $k_{d\phi} = k_{d\theta} = k_{d\psi} = 60$ and $k_{i\phi} = k_{i\theta} = k_{i\psi} = 1275.4$. For the outer-loop, we set $t_{ss}^\xi = 10$ and $\zeta^\xi = 0.99$ with $p^x = p^y = p^z = 5$.

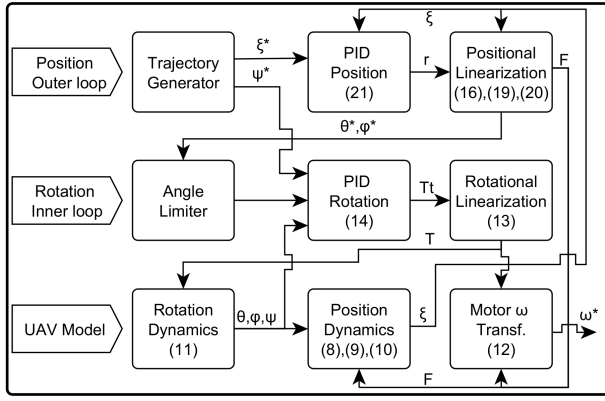


Fig. 2. Control scheme

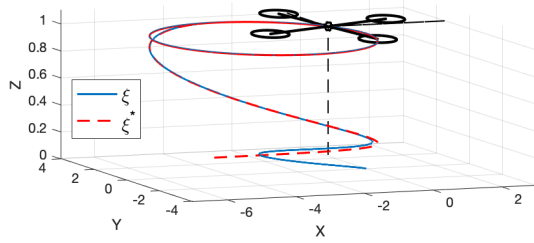


Fig. 3. Trajectory of the quadcopter in the space

Note that $t_{ss}^{\xi} \geq t_{ss}^{\eta}$ to ensure the fast convergence of the inner-loop. It generates the control gains for the posture PID $k_{px} = k_{py} = k_{pz} = 5.25$, $k_{dx} = k_{dy} = k_{dz} = 6$ and $k_{ix} = k_{iy} = k_{iz} = 1.275$. As shown in the Figure 3, the quadcopter UAV achieves the desired trajectory. The position and rotation errors in the figure 4 converge to zero. The figure 5 shows the control inputs.

IV. CONCLUSION

This paper presents the modeling of a quadcopter UAV including friction terms. The proposed control is divided in rotational inner-loop and position outer-loop using input-output linearization and PID. By doing this, we avoid the use of common truncated Taylor-series based models found in commercial flight controllers. The contribution of this paper is the calculation of the reference angles for the inner-loop using an appropriated 3D rotation matrix where the yaw angle appears. A cross-shape configuration of the UAV is considered which is preferred due to a better maneuverability than the plus-shape setup. It is ensured the fast convergence of the inner-loop respect to the outer loop according to the selection of the settling times for the PID controls. The approach becomes compatible with the commercial platforms for multi-rotor UAV's. As further research, the approach will be tested in a real UAV prototype.

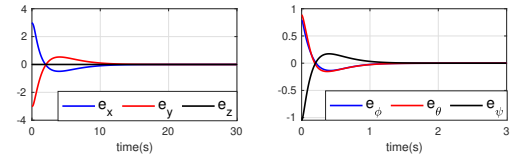


Fig. 4. Graphics of the position and rotation errors

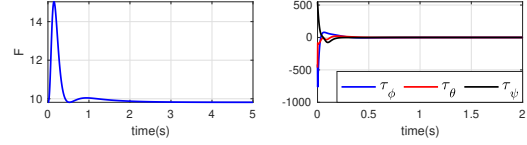


Fig. 5. Graphics of control inputs

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